

Lab 22: Method Overriding

- Due Thursday by 11:59pm
- Points 5
- Submitting an external tool
- Available Jul 21 at 6pm - Jul 23 at 11:59pm

Learning Objectives

- Practice attribute overriding
- Practice method overriding

Completion Requirements

Submit `Lab22.py`. Import the base class at the top of your file using:

```
from Lab22_provided import RentalVehicle
```

Base Class Reference: `RentalVehicle` Interface

To help you complete this lab, here is the documentation for the provided `RentalVehicle` base class. You should use these attributes and methods via `super()` where appropriate.

Attributes Initialized in `__init__(self, make, model, daily_rate)`

- `make` (str): Vehicle manufacturer (e.g., `"Ford"`).
- `model` (str): Vehicle model (e.g., `"F-150"`).
- `daily_rate` (float): Cost per day in dollars.
- `fuel_type` (str): Initialized to `"Gas"` by default.
- `min_rental_age` (int): Initialized to `21` by default.
- `maintenance_status` (str): Initialized to `"Pass"` by default.

Methods

- `apply_discount(self, percentage)`: Reduces `daily_rate` by the given percentage. Returns `None`.
- `refuel(self, amount)`: Returns the string `"Refueling with X gallons of gas."`.
- `calculate_rental_cost(self, days)`: Returns `self.daily_rate * days`.
- `can_rent(self, driver_age)`: Returns `True` if `driver_age >= self.min_rental_age`, otherwise `False`.
- `apply_mileage_charge(self, miles)`: Returns `miles * 0.25`.



- `perform_inspection(self)`: Sets `self.maintenance_status = "Pass"` and returns `"Standard safety check complete."`.
- `__str__(self)`: Returns formatted string `"Make Model - $Rate/day"`.

Q1: Electric Vehicle

Write a child class `ElectricVehicle` that inherits from `RentalVehicle`.

Attributes

Inherits all attributes from the base class, with the addition of:

- `battery_kwh` (float/int): Instance attribute representing battery capacity.
- `charge_cycles` (int): Instance attribute tracking total charge operations.

Methods

- **`__init__(self, make, model, daily_rate, battery_kwh)`:**
Call `super().__init__(make, model, daily_rate)` to initialize base attributes. Then perform the following:
 - Override `fuel_type` to `"Electric"` (instead of "Gas").
 - Override `min_rental_age` to `25` (instead of 21).
 - Initialize `self.battery_kwh` to the passed parameter.
 - Initialize `self.charge_cycles` to `0`.
- **`refuel(self, amount)`:**
Overrides the base method. It must return a string formatted as: `"Charging battery with X kWh."` (where X is `amount`) and increment `self.charge_cycles` by 1.
- **`__str__(self)`:**
Overrides the parent string method. It must call `super().__str__()` to fetch the base fleet identity string (formatted as `"Make Model - $Rate/day"`), and append the battery capacity in the exact format: `" [Battery: X kWh]"`.
- **`apply_mileage_charge(self, miles)`:**
Overrides the base method. If `miles` is 100 or fewer, return `0.0` (free). Any mileage driven *beyond* 100 miles is charged at \$0.15 per mile.

Example:

```
>> ev = ElectricVehicle("Tesla", "Model Y", 89.00, 75)
>> ev.fuel_type
'Electric'
>> ev.refuel(45)
'Charging battery with 45 kWh.'
>> ev.charge_cycles
1
>> ev = ElectricVehicle("Rivian", "R1S", 120.00, 135)
>> print(ev)
```

```
Rivian R1S - $120.00/day [Battery: 135 kWh]
>> ev = ElectricVehicle("Nissan", "Leaf", 45.00, 40)
>> ev.apply_mileage_charge(150) # First 100 free, remaining 50 miles @ $0.15
7.5
>> ev.apply_mileage_charge(80) # Under 100 miles is free
0.0
```

Q2: Truck Rental Operations

Create a new child class named `Truck` that inherits from `RentalVehicle`.

Attributes

Inherits all attributes from the base class, with the addition of:

- `cargo_tons` (float): Instance attribute representing towing capacity.

Methods

- **`__init__(self, make, model, daily_rate, cargo_tons)`:**
Call `super().__init__(make, model, daily_rate)` to initialize base attributes, and set `self.cargo_tons` to the given float parameter.
- **`calculate_rental_cost(self, days)`:**
Overrides the parent method. You must call `super().calculate_rental_cost(days)` to get the base cost first. If `days` is 7 or less, add a flat \$50 surcharge to the total. If `days` is greater than 7, print `"Surcharge Notice: Long-term discount applied to heavy-duty fee."` and add a \$20 surcharge. Return the final total cost.
- **`can_rent(self, driver_age)`:**
Overrides the parent validation method. First, check if the driver passes the base rule using `super().can_rent(driver_age)`. If the base rule passes, check an additional validation rule: if `self.cargo_tons` is greater than 5.0, the driver must be at least 25 years old. Return `True` if all conditions pass, and `False` otherwise.
- **`__str__(self)`:**
Overrides the parent string method. It must call `super().__str__()` to fetch the base fleet identity string, and append towing capacity in the exact format: `" [Cargo: X tons]"`.
- **`perform_inspection(self)`:**
Overrides the parent method. If `self.cargo_tons` is greater than 7.0, update `self.maintenance_status` to `"Requires Certified Weight Inspection"` and return `"Requires Certified Weight Inspection"`. Otherwise, return the string output returned by `super().perform_inspection()`.

Example:

```
>> heavy_truck = Truck("Ford", "F-350", 100.00, 6.2) # 6.2 tons capacity, min_rental_age is 21
>> heavy_truck.calculate_rental_cost(5)
550.0
>> heavy_truck.can_rent(22)
```

```
False
>> heavy_truck.can_rent(26)
True
>> commercial_truck = Truck("Freightliner", "M2", 250.00, 8.0)
>> print(commercial_truck)
Freightliner M2 - $250.00/day [Cargo: 8.0 tons]
>> super_truck = Truck("Volvo", "VHD", 350.00, 15.0)
>> super_truck.perform_inspection()
'Requires Certified Weight Inspection'
```

This tool needs to be loaded in a new browser window

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